# YING ZHANG

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## EDUCATION

**UNIVERSITY OF CAMBRIDGE** Cambridge, UK
**MPhil in Data Intensive Science, Lucy Cavendish College** *Oct 2026 - July 2027*
**UNIVERSITY OF NOTTINGHAM** Nottingham, UK
***Bachelor of Science (Honours), Statistics - First Class*** *Oct 2022 - June 2026*
***Major Courses:*** *Complex Analysis, Differential Equations, Vector Calculus and Electromagnetism, Real Analysis, Scientific Computation, Foundation of Pure Maths, Applied Mathematics, Analytical Computing,Statistical Inference (**97**), Probability and Statistics (**90**), Complex Analysis (**89**)*

## RESEARCH EXPERIENCE

**Compressed Latent Diffusion Language Models (CLDLM)** Hangzhou, CN
**Research Intern, ZJU-UIUC, co-author manuscript under review, NeurIPS 2026 Workshop** *Jul 2026 – Sep 2026*

- Addressed length-compressed diffusion language modeling, where prior two-stage pipelines fix an autoencoder before training the generative model, by jointly training a compressor, a Flow Matching model, and a token decoder end to end so the compressed representation adapts to generation.
- Coupled JEPA-style joint-embedding prediction with Flow Matching by using the Flow Matching model itself as the predictor at the final noise step, so one network both predicts and denoises the compressed embedding during generation.
- Benchmarked against seven autoregressive, discrete-diffusion, and flow-based baselines including ELF on LM1B and OpenWebText with five-seed evaluation, achieving the lowest generation perplexity and highest throughput among all diffusion and flow models, 26.20 Gen-PPL on OpenWebText at 2.4x ELF's throughput (10,511 vs. 4,416 tokens/s).

**Klein Neural Networks++: Closed-Form Layers on the Klein Model of Hyperbolic Space** Trento, Italy
**Research Intern, University of Trento, First-author manuscript under review, NeurIPS 2026** *Oct 2025 – April 2026*

- Derived a closed-form Klein batch normalisation that replaces the iterative Fréchet mean required by prior gyrogroup BN with a single-pass Einsteinmidpoint, removing the inner optimisation loop at negligible accuracy cost.
- Developed the first systematic set of intrinsic Klein layers, spanning classification, fully connected, and normalisation heads in both geodesic and horosphere form, from a single Klein hyperplane distance. The model had stayed unexplored while the field standardised on the Poincaré ball and Lorentz hyperboloid, despite being the only one of the three admitting both Euclidean geodesics and a closed-form midpoint.
- Benchmarked against Poincaré, Lorentz, and PV counterparts on image classification (CIFAR-10/100, Tiny-ImageNet, ImageNet-1k), graph link prediction, and node classification, where Klein matches or improves on all three while remaining stable under feature clipping, curvature mis-specification, and gradient scaling

**When Early Is Not Early: The Precursor Illusion in Time-Series Anomaly Detection** Beijing, CN
**Research Intern, Tsinghua University, Co-first-author manuscript under review, AAAI 2027** *Apr 2026 – Aug 2026*

- Formalised the precursor illusion where pre-onset labels are defined by temporal position alone and point adjustment is applied, causing the reactive detectors to appear to deliver advance warning they never had.
- Designed a model-agnostic, placebo-calibrated test for whether pre-onset intervals actually contain detectable evidence, finding fewer than one in seven benchmark events qualify once the placebo baseline is subtracted.
- Showed that after multiplicity correction none of 35 model-dataset cells across 7 detectors discriminates precursor from matched normal windows, while point adjustment inflates mean detection F1 from 0.186 to 0.549 with scores and thresholds held fixed.

**Simulation Based Amortised Inference (SBI) using Neural Posterior Estimation** (github repo) Nottingham, UK
**Research Intern funded by Engineering and Physical Sciences Research Council (EPSRC)** *Jun 2025 – Aug 2025*

- Built a 3-layer DNN for simulation-based inference integrating normalizing flows, sufficient statistics, and CLT structural regularization, simulated gamma & beta datasets, benchmarked mixture of gaussians vs. normalizing flow
- Designed conditional normalizing flow architectures for i.i.d. *Poisson* models, achieved accurate inference of non-analytic posteriors in large-N regimes, proved gaussian-alignment improved stability over standard neural inference
- Presented project outcomes at the University of Nottingham Undergraduate Research Poster Exhibition (Nov 2025), highlighting multi-distribution benchmarking and novel CLT-based regularization to an audience of faculty and peers

## PUBLICATIONS AND MANUSCRIPTS

- **Ying Zhang**, Nicu Sebe, Bernhard Schölkopf, Ziheng Chen, *Klein Neural Networks++,* NeurIPS 2026, Under Review (first author)
- Yue Yang*, **Ying Zhang***, Yiwen Yan, Chen Zhang, *When Early Is Not Early: The Precursor Illusion in Time-Series Anomaly Detection,* AAAI 2027, Under Review (*equal contribution)
- Yulin Yuan, **Ying Zhang**, Xiangming Meng, *One Latent, Many Tokens: Jointly Learning Compressed Representations for Efficient Language Diffusion,* NeurIPS 2026 Workshop, Under Review
- Yue Yang, Zihan Su, **Ying Zhang**, Chang Chuan, Yuxiang Lin, Anthony Graham Bellotti, Boon Giin Lee, *Kolmogorov–Arnold Networks-based GRU and LSTM for Loan Default Early Prediction,* Journal of Applied Soft Computing 2025, Accepted (arXiv:2507.13685)
- Yue Yang, Yuxiang Lin, **Ying Zhang**, Zihan Su, Anthony Graham Bellotti, Boon Giin Lee, *Transforming Credit Risk Analysis: A Time-Series-Driven ResE-BiLSTM Framework for Post-Loan Default Detection, The Third Credit Scoring and Credit Rating Conference (CSCR III)*, 2024, Accepted (arXiv:2508.00415)

## WORK EXPERIENCE

**Trendy Group (Ochirly, Miss Sixty & Five Plus)** (page) Guangzhou, CN
**Data Analyst Intern** *Jun – Aug 2024*

- Designed, developed, and tested a rolling inventory system for Five Plus using IBM Planning Analytics, integrating master business data for automated calculations, which improved product turnover and reduced inventory losses by 10%
- Built an OTB ordering tag system leveraging multidimensional database analytics to optimize annual order quantities based on sales targets, strategy, and historical performance, replacing manual Excel workflows, cut calculation time
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## TECHNICAL SKILLS

- **Programming:** Python (PyTorch, NumPy, SciPy, scikit-learn, geoopt), R, MATLAB, LaTeX, Git
- **Research Computing:** Slurm job arrays and multi-GPU training on Ada (Nottingham) and CINECA Leonardo (EuroHPC), plus mixed-precision and numerical-stability debugging
- **Methods:** Hyperbolic and Riemannian deep learning, normalizing flows, diffusion and score-based models, simulation-based inference, Bayesian inference, time-series anomaly detection

## HONORS AND AWARDS

- Mathematical Contest in Modeling, National 1st Prize of the Year 2023 (github repo, paper)
- Shuwei Cup of Mathematical Modeling Challenge, National 2nd Prize of the Year 2023 (github repo, paper)
- National Mathematical Modeling Competition, National 3rd Prize of the Year 2023 (github repo, paper)
- Huashu Cup of Mathematical Contest in Modeling, National 3rd Prize of the Year 2023 (github repo, paper)
- Mathematical Contest in Modeling, Successful Prize of the Year 2024 (github repo, paper)
- Mathematical Contest in Modeling, Honorable Prize of the Year 2025 (github repo, paper)